

Trigram HMM

Smoothed Estimation

The transition parameters can be smoothed just like n-gram models

$$q(Vt \mid DT, JJ) = \lambda_1 \times \frac{\text{Count}(Dt, JJ, Vt)}{\text{Count}(Dt, JJ)} \\ + \lambda_2 \times \frac{\text{Count}(JJ, Vt)}{\text{Count}(JJ)} \\ + \lambda_3 \times \frac{\text{Count}(Vt)}{\text{Count}()}$$

$$\lambda_1 + \lambda_2 + \lambda_3 = 1, \quad \text{and for all } i, \lambda_i \geq 0$$

For emission probabilities we usually use pseudowords

$$e(\text{base} \mid Vt) = \frac{\text{Count}(Vt, \text{base})}{\text{Count}(Vt)}$$

HMM: Smoothing with Pseudowords

Pseudowords use our existing information about the task to categorize words that appeared very few times, or not at all, in the training data

A common method is as follows:

- ▶ **Step 1:** Split vocabulary into two sets

Frequent words = words occurring ≥ 5 times in training

Low frequency words = all other words

- ▶ **Step 2:** Map low frequency words into a small, finite set, depending on prefixes, suffixes etc.

HMM: Smoothing with Pseudowords

[[Bikel et. al 1999](#)] (named-entity recognition)

Word class	Example	Intuition
twoDigitNum	90	Two digit year
fourDigitNum	1990	Four digit year
containsDigitAndAlpha	A8956-67	Product code
containsDigitAndDash	09-96	Date
containsDigitAndSlash	11/9/89	Date
containsDigitAndComma	23,000.00	Monetary amount
containsDigitAndPeriod	1.00	Monetary amount, percentage
othernum	456789	Other number
allCaps	BBN	Organization
capPeriod	M.	Person name initial
firstWord	first word of sentence	no useful capitalization information
initCap	Sally	Capitalized word
lowercase	can	Uncapitalized word
other	,	Punctuation marks, all other words

HMM: Smoothing with Pseudowords

Profits/NA soared/NA at/NA Boeing/SC Co./CC ,/NA easily/NA
topping/NA forecasts/NA on/NA Wall/SL Street/CL ,/NA as/NA their/NA
CEO/NA Alan/SP Mulally/CP announced/NA first/NA quarter/NA
results/NA ./NA



firstword/NA soared/NA at/NA initCap/SC Co./CC ,/NA easily/NA
lowercase/NA forecasts/NA on/NA initCap/SL Street/CL ,/NA as/NA
their/NA CEO/NA Alan/SP initCap/CP announced/NA first/NA
quarter/NA results/NA ./NA

NA = No entity
SC = Start Company
CC = Continue Company
SL = Start Location
CL = Continue Location